



Bizz Summit
ES 2023

Building Time Tracking solution with Power Platform

Timo Pertilä

PATROCINADORES BIZZ SUMMIT ES 2023

PLATINO



minsoit

An Indra company

esic



avanade



V-Valley



sogeti
Part of Capgemini

infoAVAN

ORO



axazure
by avvale

RICOH
imagine. change.

intelequia

TOKIOTA

prisma

NTT DATA



devoteam



bravent

VSSistemas



adderit



nunsys

encamina
PIENSA EN COLORES

prodware



sothis

ECM

ENTERPRISE
QUALITY
MANAGEMENT

PLATA



softeng
Maximize the cloud.
Securely.



CrossPoint

isolutions

En colaboración con:



Microsoft



TIMO PERTILÄ

Power Platform consultant, blogger, slow runner

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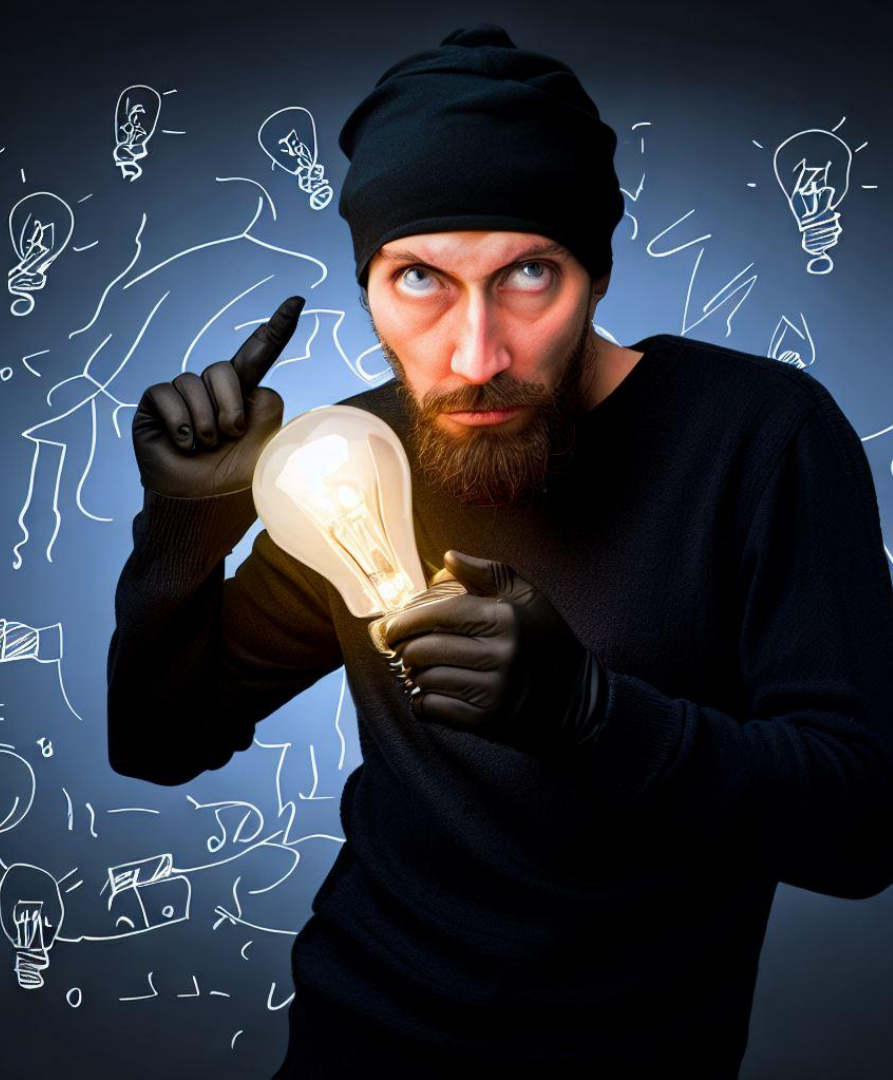


2020



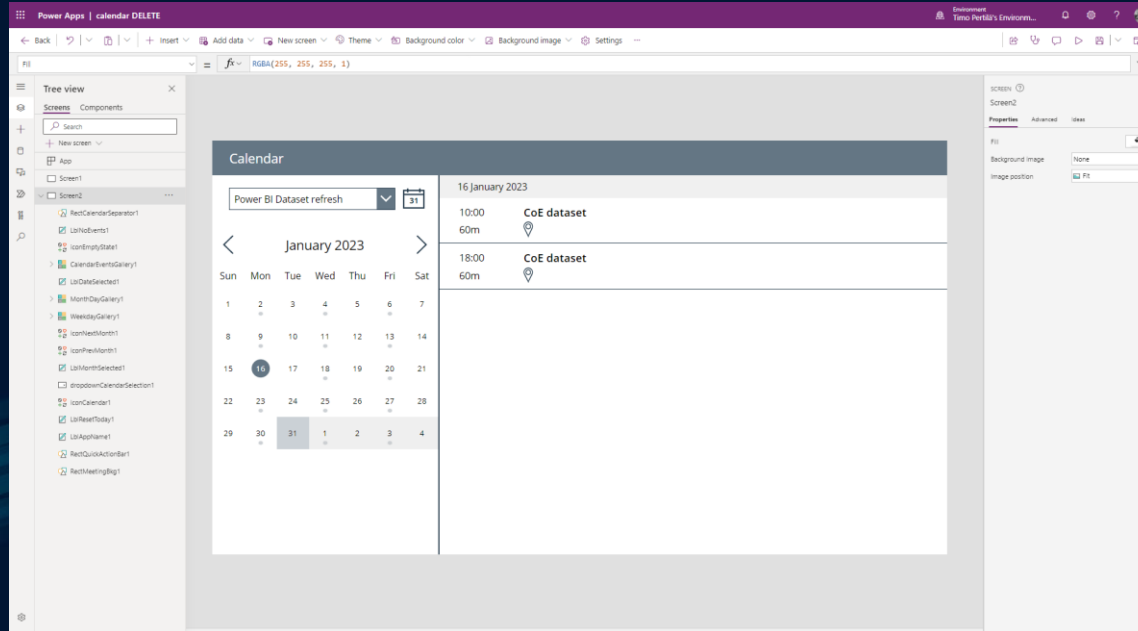
03/2020

- Starting a new company, hour tracking solution needed
- Readymade product vs Power Platform
- Accounts and opportunities already stored in Dynamics 365
- “Just add”
 - Work orders
 - Hour entries
 - App
- Implement only what we **really** need. Everything else when we know better what we need / have more time
- “Who will build this? Ok, Timo”



**Steal whenever
you can**

STARTING POINT - CALENDAR EVENTS SCREEN



Changes

- Replace calendar events with hour entries
- Add form for add / modify hour entries
- Monthly and weekly totals shown somewhere

FIRST WORKING VERSION

Work hours for customers we love

Tuesday, March 24, 2020

<

March 2020

>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	1	2	3	4

Selected week

0h

Selected month

21.5h

Timon työt maaliskuu
kehityskäytännöt

1h

>

Add work hour

* Customer

Find items

* Work Order

Find items

* Hours

Description

* Day

24.3.2020

Save

LIGHT MAKEOVER

FORWARD
FOREVER

Work hours for customers we love

Tuesday, March 3, 2020

Selected month
29.25 h

Selected week
2.75 h

Selected day
2.75 h

< March 2020 >

Mon Tue Wed Thu Fri Sat Sun

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

Timon työt maaliskuu

2.75h >

Modify work hour

Customer

Work Order

Timon työt maaliskuu

Hours

2.75

Description

Date

3.3.2020

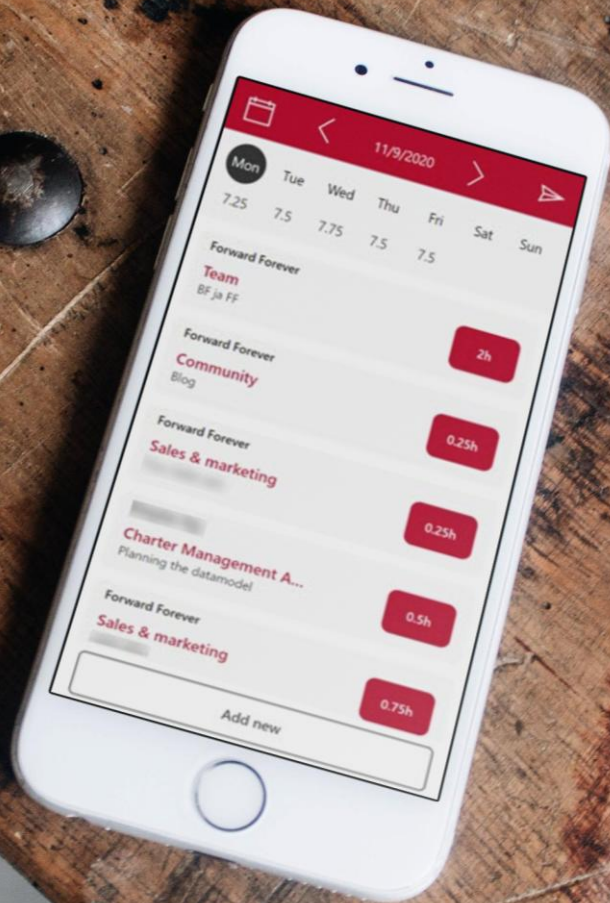
Cancel

Save


Bizz Summit
ES 2023

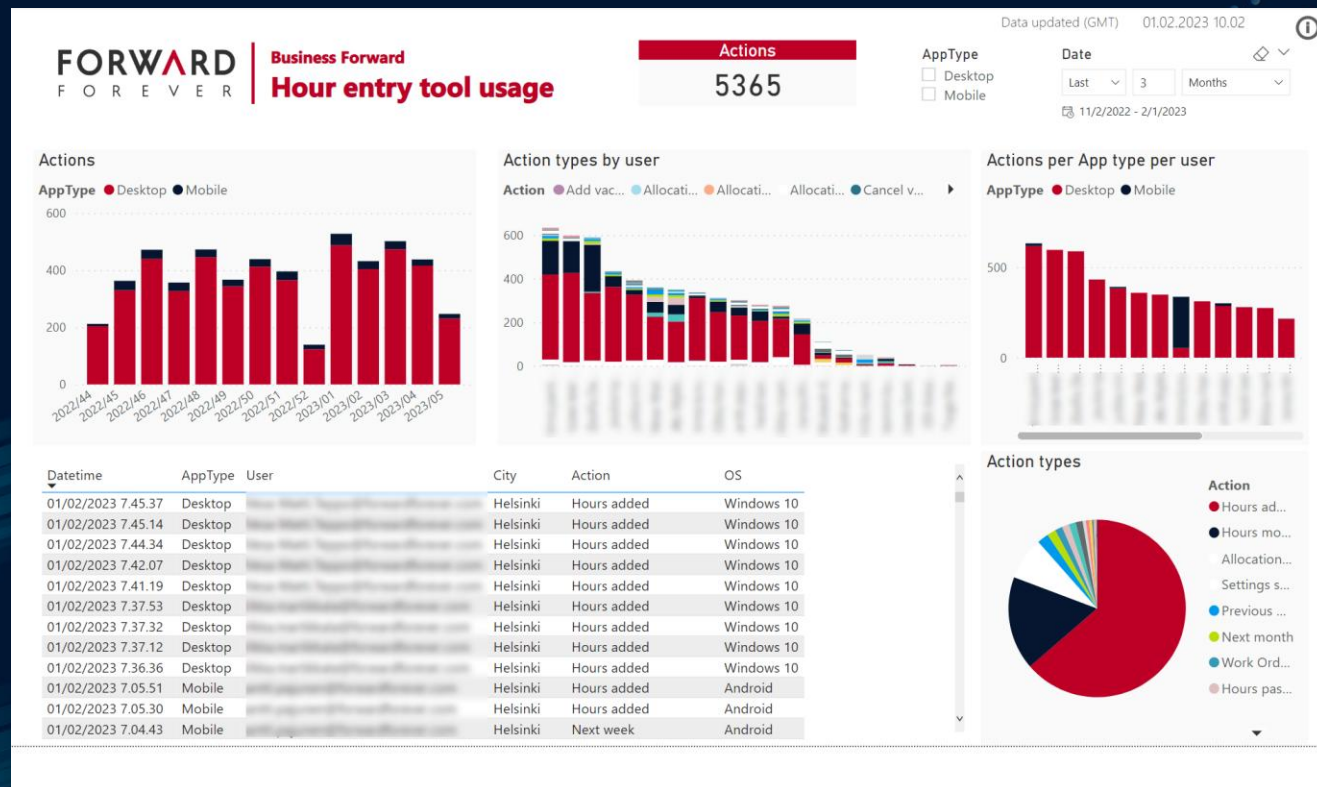
An ornate, gold-colored frame with intricate scrollwork and floral patterns surrounds the text. The frame is set against a textured, light brown background.

Make it work
Make it pretty
Make it right



At the same
time...
Mobile App

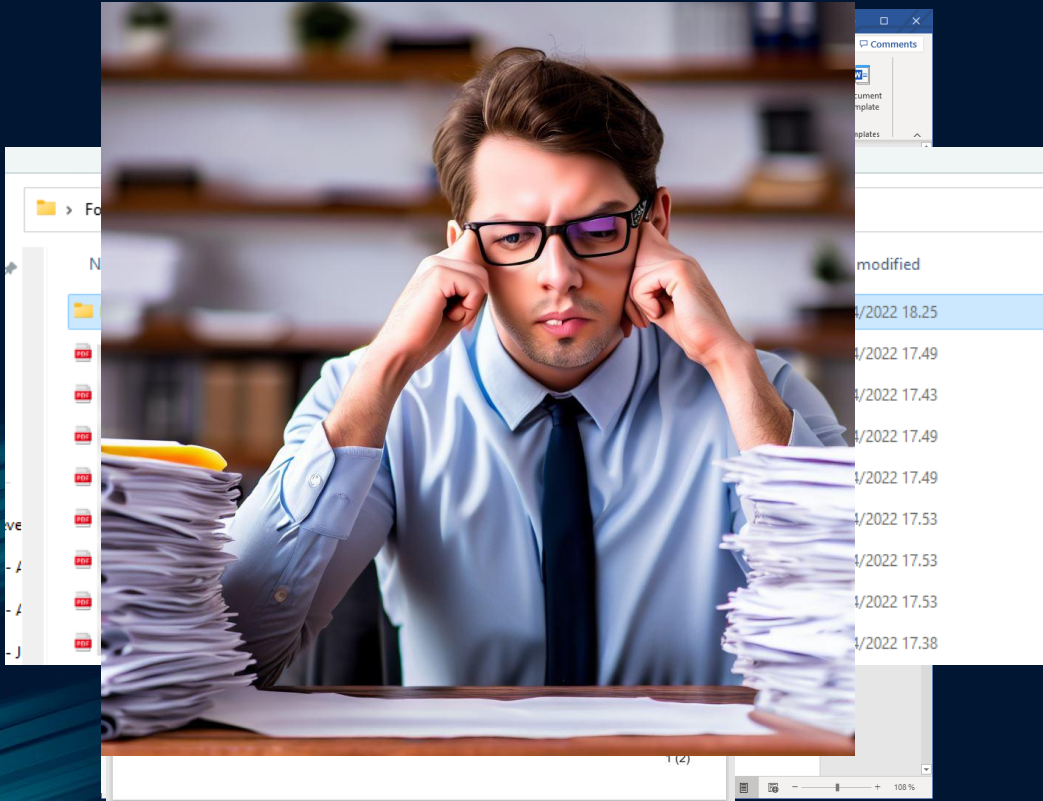
USAGE ANALYTICS - APPLICATION INSIGHTS





**Finally,
development
starts**

INVOICE ATTACHMENTS



- Manually created invoice attachments generated -> Time consuming
- Decided to automate attachment generation (April 2020)
- Attachments created with flow and stored into the SharePoint document library
- No changes to Hour Entry App

2021



PERFORMANCE IMPROVEMENTS

In January 2021 users started to claim about slowness

- Query optimization
- Collection optimization
- Dropped out company level month/week/day summary

PERFORMANCE IMPROVEMENTS

FORWARD
FOREVER

Work hours for customers we love

Tuesday, March 3, 2020

Selected month

29.25 h

Selected week

2.75 h

Selected day

2.75 h

<

March
2020

>

Mon	Tue	Wed	Thu	Fri	Sat	Sun
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Timon työt maaliskuu

2.75h

>

Modify work hour

Customer

Work Order

Timon työt maaliskuu

Hours

21.5 h done by 3/26/2020 1:29 PM

2.75

Description

Date

3.3.2020

Cancel

Save

EXPENSES

- Expenses were handled with SharePoint document library until January 2021
- Added expenses feature into the mobile app
 - User see expense statuses in the app
 - Used AI Builder for reading expense information
 - Expenses still stored into the SharePoint
 - AI Builder parsed value for 10% and Vendor for 32% of expenses
- Feature dropped in 2022 (replaced with Revolut cards)

Expense details

+ New Upload Edit in grid view Sync Add shortcut to OneDrive Pin to Quick access Export to Excel Automate Integrate

Expenses PAID

Name	Total value	ValueFromAI	Vendor	VendorFromAI	Description	Paid
	12		K-Market Kamppi		toimistolle	
	37,12	✓	K-supermarket Kamppi	✓	Välipalokkeja toimistolle	✓
	20,86	✓	K-supermarket Kamppi	✓	Braseroa ja välipalokkeja	✓
	18,94	✓	K-supermarket Kamppi	✓	Välipalokkeja ja limua	✓
	26,95	✓	K-supermarket Kamppi	✓	Kaffea ja Braseroa	✓
	4,99	✓	K-supermarket Kamppi	✓	Huipattareita	✓
	27,8	✓	POBRE	✓	Tönnin eläin päivän lounas	✓
	24,8	✓	DNA	✓	Prepaid 4G 30d toimiston varanetti	✓
	52,07	✓	ALERA KAMPINKESKUS	✓	FF vapputarjoilut	✓
	656	✓	BASBAS KULMA COUNTER & DINERY	✓	FF tv tarjoilut	✓
	53,01	✓	KÄLVIKKÄVERIUS OH/ MEB	✓	MEB paita	✓
	81	✓	Lasipalatin Laituri	✓	Lasipalatin tarjoilut - FF vappu	✓
	90	✓	Master Golf Course Oy Ab	✓	Microsoft Partner Fiscal Year Co-Planning golf event	✓

Cancel Save

MY WORK ORDERS

- In February 2021 there was so many customers and work orders that it slowed down adding hour entries
- My Favourites –feature added after short discussion

The screenshot displays the 'Add work hour' interface of a mobile application. On the left side, there is a vertical list of time entry options: '1h', '4h', '0.5h', and '2h', each with a right-pointing arrow. The main area is a form titled 'Add work hour'. Inside the form, there is a 'Customer' field, followed by a 'Find items' section. The 'Find items' section contains a list of blurred items, and a red arrow points to the 'Find items' header. At the bottom right of the form is a 'Save' button. On the far right, there is a red button labeled '2.75h' with a right-pointing arrow.

MANAGING MY FAVOURITES

FORWARD
FOREVER

My Favourite Work Orders

Back

Work orders

Search with customer or work order

Very Important Customer
App for Social Distancing

Add

Very Important Customer
Governance in practice

Add

Test Account 1
Governance Starter Kit

Add

Very Important Customer
Important work

Add

Test Account 1
Monthly support

Add

Fake Corp LTD
Potato Invoicing

Add

Customer 1
Power Apps Consulting

Add

Customer 2
Power Apps Project - Tool 1

Add

Customer 3
Power BI Consulting

Add

My work orders (6)

Search with customer or work order

Fake Corp LTD
Potato Invoicing

Test Account 1
Governance Starter Kit

Test Account 1
Monthly support

Test Account 1
Proof of Concept

Very Important Customer
App for Social Distancing

Very Important Customer
Important work

USING MY FAVOURITES



FORWARD
FOREVER

Work hours for customers we love

13 September 2022 

My hours

Selected month
60 h

Selected week
15 h

Selected day
7.5 h

< September 2022 >

Mon	Tue	Wed	Thu	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

Customer 3

Power BI Consulting
dsdas

3h >

Global Corporation Ltd

Small extra work
dsa

1.5h >

Customer 1

Power Apps Consulting
dsa

3h >

Add work hour

Customer

Customer 1 ✓

Work Order

Find items

Power Apps Consulting

Power Platform Advisory Service

Description

Date

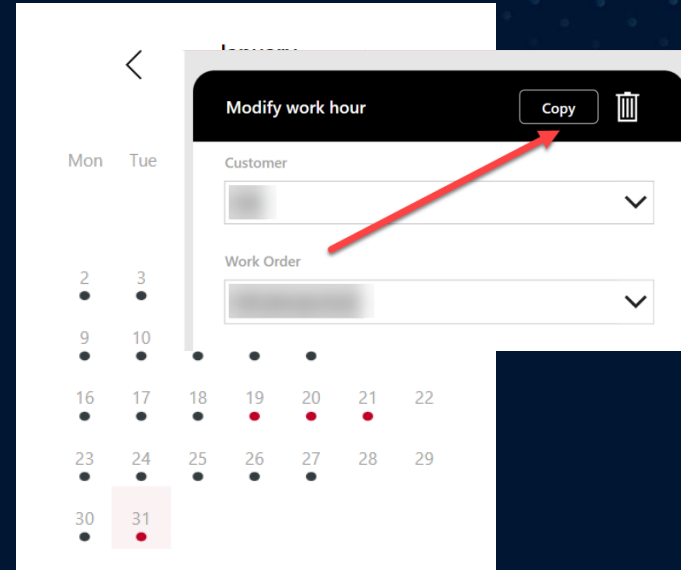
13.9.2022 

Save



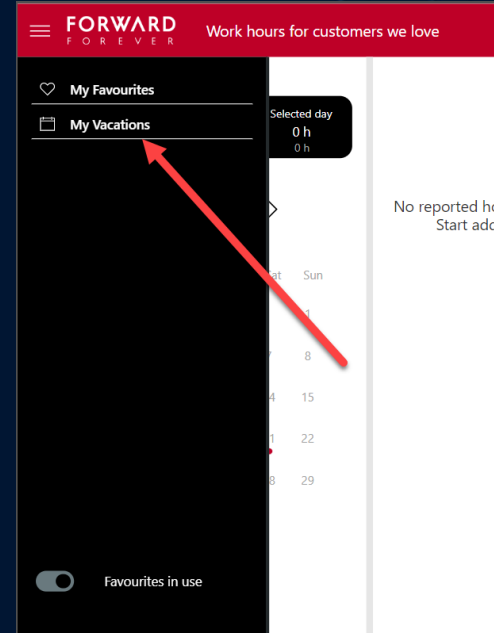
OTHER SMALL IMPROVEMENTS

- Partially reported days marked with red dot
- Copy-pasting hour entries



VACATIONS

- After winter holidays I decided to automate some part of vacation related routines



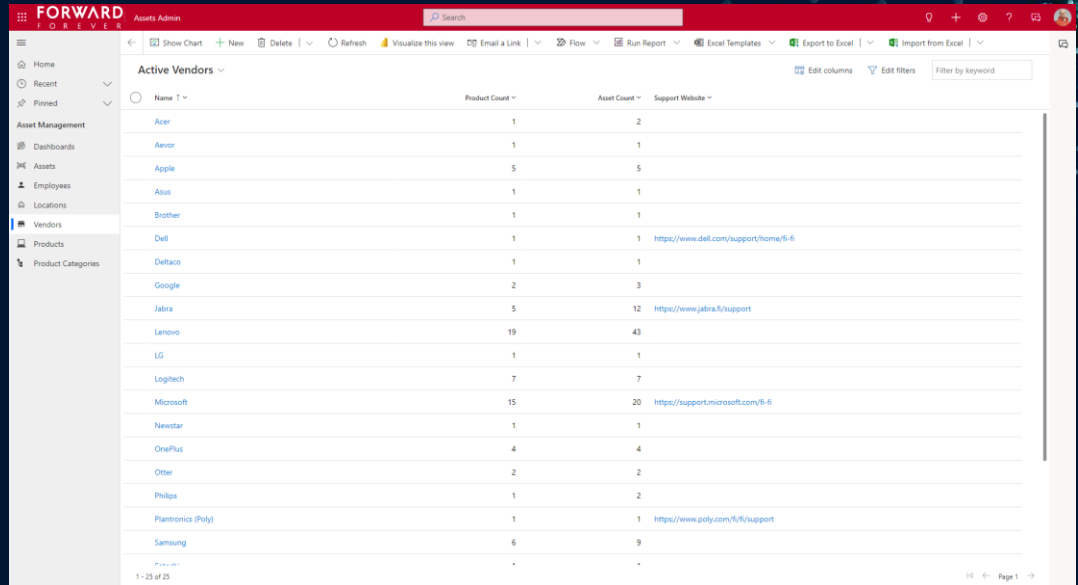
akes hour entries for the
vacation period
dds vacation to the shared
vacation calendar
dds vacation into my Outlook
calendar

Tool

- Makes hour entries for the vacation period
- Adds vacation to the shared vacation calendar
- Adds vacation into my Outlook calendar
- Adds out of the office message

ASSET MANAGEMENT

- In April we realized that we need an asset management system

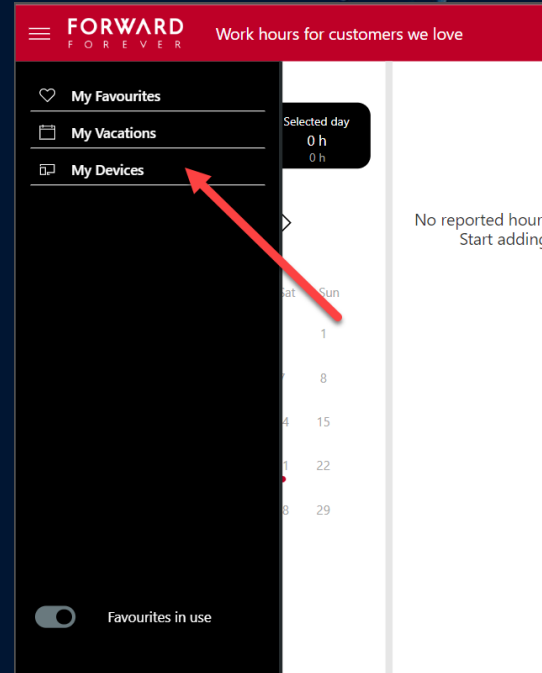


The screenshot displays the 'FORWARD FOREVER Assets Admin' web application. The interface includes a sidebar with navigation options: Home, Recent, Pinned, Asset Management, Dashboards, Assets, Employees, Locations, Vendors (selected), Products, and Product Categories. The main content area shows a table titled 'Active Vendors' with columns for Name, Product Count, Asset Count, and Support Website. The table lists various vendors and their corresponding counts. At the bottom of the table, it indicates '1 - 25 of 25' items.

Name	Product Count	Asset Count	Support Website
Acer	1	2	
Aenvor	1	1	
Apple	5	5	
Asus	1	1	
Brother	1	1	
Dell	1	1	https://www.dell.com/support/home/%66
Dellaco	1	1	
Google	2	3	
Jabra	5	12	https://www.jabra.fi/support
Lenovo	19	43	
LG	1	1	
Logitech	7	7	
Microsoft	15	20	https://support.microsoft.com/%66
Newstar	1	1	
OnePlus	4	4	
Otter	2	2	
Philips	1	2	
Plantronics (Poly)	1	1	https://www.poly.com/%66/support
Samsung	6	9	

MY DEVICES

- Employee see all her/his assets in Hour Entry App



MY DEVICES

FORWARD

FOREVER

My Devices

Add Device

★ ★ ★ ★ ★

iPhone 12 Pro 128GB

Apple

Serial No. 355942853072076

★ ★ ★ ★ ★

Samsung F27T850Q 27"

Samsung

Serial No.

★ ★ ★ ★ ★

Surface Precision

Microsoft

Serial No.

★ ★ ★ ★ ★

Surface Laptop 3 13.5", i7, 16GB

Microsoft

Serial No. A0ADF0BC-C965-4533-BB69-B3921EB0D18A

★ ★ ★ ★ ★

Lenovo ThinkVision P27h-10

Lenovo

Serial No.

Modify Device

Product

Surface Precision

Add

Purchased From

Gigantti

Purchase Date

9/25/2020

Serial Number

Model Number

Warranty End Date

12/31/2001

Comments

Cancel

Save


Bizz Summit
ES 2023

PERSONAL HOUR BALANCE

FORWARD

FOREVER

Work hours for customers we love

Wednesday, February 1, 2023

My hours

Selected month

0 h

0 h

Selected week

7.5 h

14 h

Selected day

0 h

0 h

<

February

>

2023

Mon	Tue	Wed	Thu	Fri	Sat	Sun	
		31	1	2	3	4	5
6	7	8	9	10	11	12	
13	14	15	16	17	18	19	
20	21	22	23	24	25	26	
27	28						

Work hour balance

12.25

No reported hours for Wednesday, February 1, 2023.
Start adding hours with form on right.

Add work hour

Customer

Forward Forever

Work Order

Find items

Hours

Description

Date

1.2.2023

Save



2022

ALLOCATIONS

- Employees asked for tool to plan allocations
- Already tried with shared Excel file (everyone hated)

Excel FF Resursointi

Max workdays per month: ~20

80% 16

60% 12

Summa / Työmäärä (htp)

	2021/01	2021/02	2021/03	2021/04	2021/05	2021/06
Jouko Nyholm	14	28	20	13	27	21
Timo Perttälä	16	20	21	17	26	22
		1				
	5	10	4	5	4	4
	2	2	6	5	2	2
			1			1
	1					
	2	2	1	1	4	
		2	1			
	3	1	3	3	3	1
		1				
	2	1				1
	1					
			4	0	0	
			1	1	6	7
			1	1	1	1
			1	5	5	
Jukka Nilanen	10	16	17	14	27	20
Lasse Teeriäho	26	25	20	16	14	19
Antti Pajunen	9	12	1,5	11	12	7
Hilla Mäntymäen	21	21	6	16	8	3
Heidi Korhonen		15	20	5	12	12
Tommi Kurula					7	6
Timo Turunen					5	6
Grand Total	96	137	105,5	92	138	116

Kausi

Asiakas

MANAGING MY ALLOCATIONS

FORWARD
FOREVER

My work for next months

Back

Search existing work orders

Planning is done on work day level

Add new work order

Oct/2022

Nov/2022

Dec/2022

Jan/2023

<

Jan/2023

Feb/2023

Mar/2023

Apr/2023

>

15.2

23.2

17.1

21.1

16.5

20

14

12.5

	Customer 1	Power Platform Advisory Servi...	0.1	3	0.5	2				
	Customer 1	Power Apps Consulting	0	0.5	0.1	0.6	1.5	1.5	1.5	0.5
	Forward Forever	Team	1.6	2.3	2	1.1	1	1	1	1
	Forward Forever Oy	Internal Project X						5		
	Forward Forever Oy	Power Platform Support	0.7	0.8	0.4	2.1	1	1	1	1
	test account	FF		1	1.2	6	3	3	3	3
	Test Account 1	Governance Starter Kit		2.8	1.3	1.2	3	3	3	3

Changes: 0

Save

OTHER SMALL IMPROVEMENTS

- Support for different working times (7.5h vs 8h per day)
- Navigation to the Work Order page in D365
- Budget vs actual

The screenshot shows the 'Modify work hour' form in D365. At the top, there is a header bar with the title 'Modify work hour' and two buttons: 'Copy' and a trash icon. Below the header, the form contains several fields:

- Customer:** A dropdown menu with a blurred selection and a downward arrow.
- Work Order:** A dropdown menu with a blurred selection and a downward arrow. A red arrow points to a small icon (a document with a checkmark) to the right of the dropdown.
- Hours:** A text input field containing '0.5'. Above the input field, the text '54.25/86.25 h done by 30.1.2023' is displayed. A red arrow points to this text.
- Description:** A text area containing the text 'Määrittely, tietomalli, alustava ratkaisu'.
- Date:** A text input field containing '16.12.2022' and a calendar icon to its right.

At the bottom of the form, there are two buttons: 'Cancel' and 'Save'.

2023

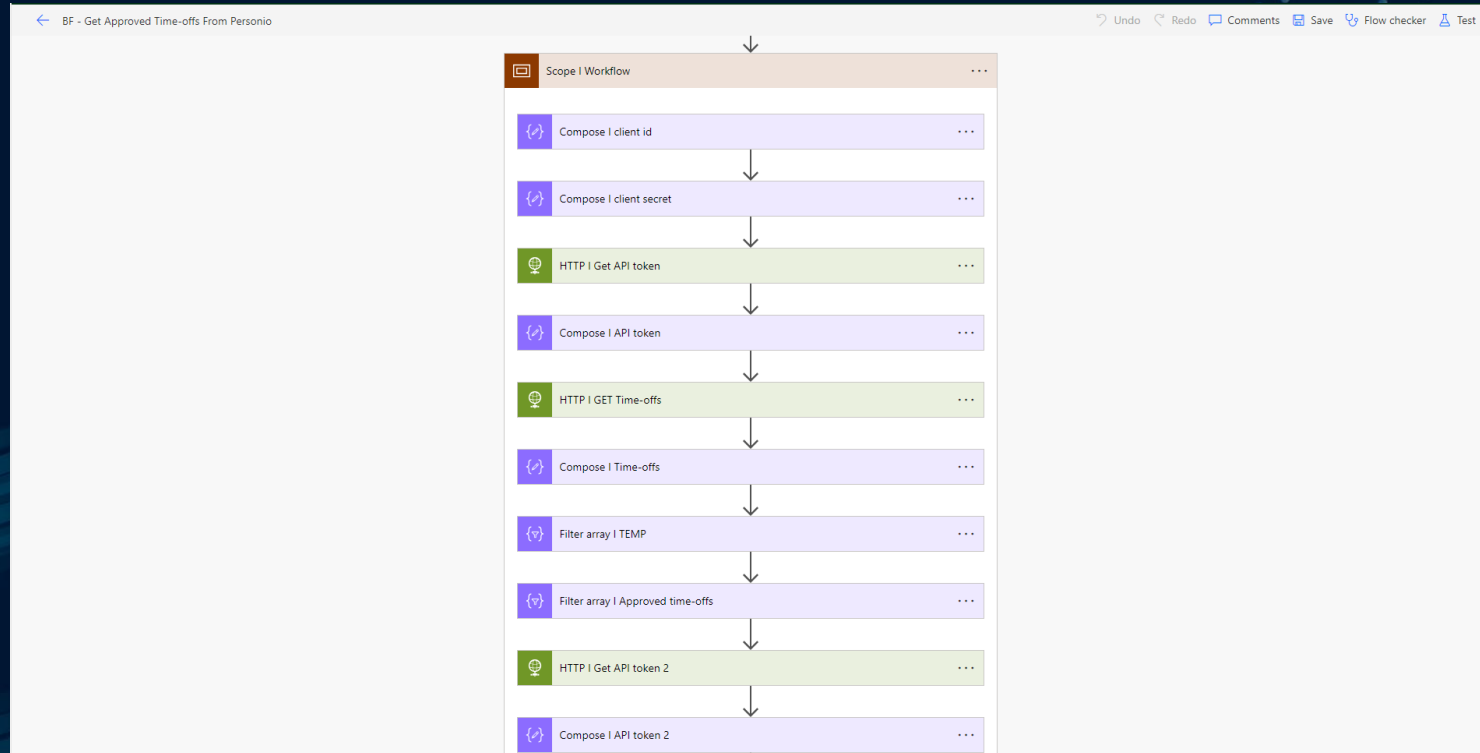


PERSONIO INTEGRATION

The screenshot displays the FORWARD FORUM application interface. A modal titled "Request Absence" is open in the center. The background interface includes a sidebar with navigation links: Home, My profile, Employees, and Calendars. The main content area shows the user's profile for Timo Pertilä, with sections for Tasks (0), Reminders, Out today, and Birthdays. The "Request Absence" modal contains the following fields:

- What type of absence?**: A dropdown menu with "Annual leave (vuosiloma in Finnish)" selected.
- From - To**: Two date input fields with the format DD/MM/YYYY.
- Substitute (Optional)**: A dropdown menu with the text "Only people who are working during this time will appear in the list." and "Choose a substitute..." below it.
- Comment (Optional)**: A text input field with the placeholder "Add a comment..." and a small icon on the right.
- Buttons**: "Cancel" and "Request" buttons at the bottom.

PERSONIO INTEGRATION



PERSONIO INTEGRATION

FORWARD

FOREVER

Work hours for customers we love

11 July 2023 

My hours

Selected month

0 h

150 h

Selected week

0 h

37.5 h

Selected day

0 h

7.5 h

<

July

>

2023

Mon

Tue

Wed

Thu

Fri

Sat

Sun

1

2

3

4

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10

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12

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21

22

23

24

25

26

27

28

29

30

31

Work hour balance

35.25

Forward Forever

Vacation

Approved absence from Personio

7.5h



Add work hour

Customer

Find items

Work Order

Find items

Hours

Description

Date

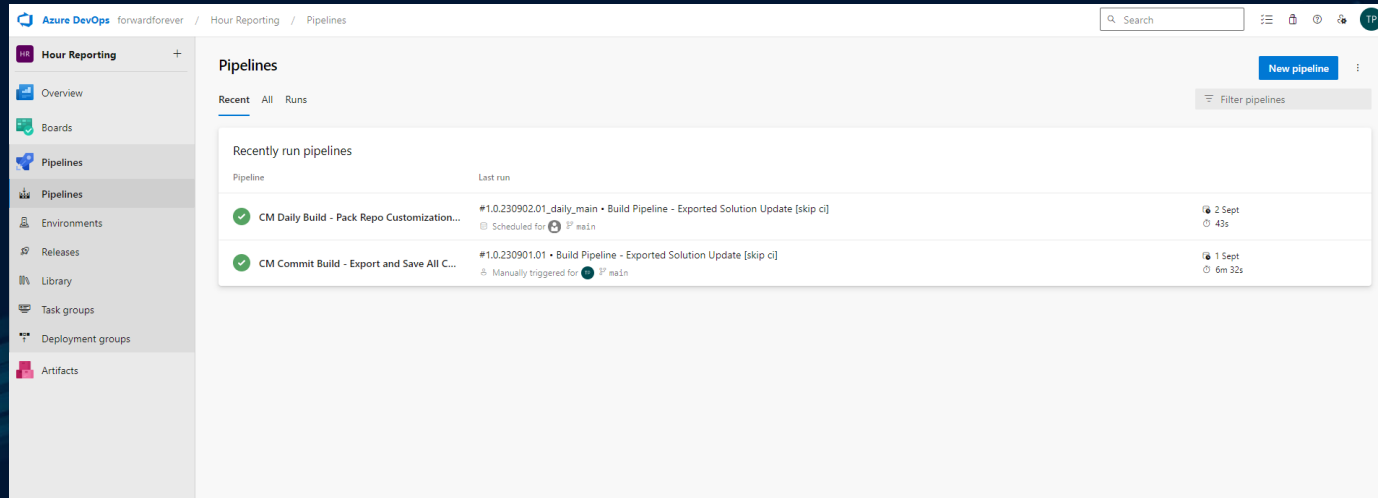
11.7.2023



Save


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DEPLOYMENTS WITH AZURE DEVOPS



The screenshot displays the Azure DevOps web interface. The top navigation bar shows the user 'forwardforever' and the current path 'Hour Reporting / Pipelines'. A search bar is located on the right. The left sidebar contains a menu with options: Overview, Boards, Pipelines (selected), Environments, Releases, Library, Task groups, Deployment groups, and Artifacts. The main content area is titled 'Pipelines' and has tabs for 'Recent', 'All', and 'Runs'. A 'New pipeline' button is in the top right. Below the tabs, a 'Filter pipelines' button is visible. The 'Recently run pipelines' section contains a table with two entries:

Pipeline	Last run
✓ CM Daily Build - Pack Repo Customization...	#1.0.230902.01_daily_main • Build Pipeline - Exported Solution Update [skip ci] ⌚ Scheduled for 3P maIn ⌚ 2 Sept ⌚ 43s
✓ CM Commit Build - Export and Save All C...	#1.0.230901.01 • Build Pipeline - Exported Solution Update [skip ci] ⚙ Manually triggered for 3P maIn ⌚ 1 Sept ⌚ 6m 32s

NEW COMPANY BRAND

- Total makeover when new brand was released
- Fully responsive app, no more dedicated mobile version
- Some of rarely used features were left out

Calendar

Power BI Dataset refresh

▼

20

<

September 2023

>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

Monday August 28, 2023

10:00 AM

60m

CoE dataset



6:00 PM

60m

CoE dataset



[Home](#) [Allocations](#) [Favourites](#) [My devices](#)

My hours

Month	Week	Day
7 h	7 h	7 h
123 h	40.5 h	12 h

< September 2023 >

Mon	Tue	Wed	Thu	Fri	Sat	Sun
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

Thursday, 21 Sep (12 h)

Customer 1 1h

Power Apps Consulting
All kind of small stuff

Very important Customer 2h

Governance in practice
Whoaaaaaa

Very Important Customer 1h

App for Social Distancing
Long description here

Forward Forever 5h

Team

Forward Forever Oy 3h

Power Apps Project - Tool 1
Description here

Add work hour

Customer

Find items

Work Order

Find items

Hours

Description

Date

21.9.2023

Save

OUR TIME TRACKING SOLUTION IN A NUTSHELL

- Dataverse based canvas app for keeping track of work hours
- Reporting with Power BI
- Usage analytics with Application Insight + Power BI
- Built on dev environment, manually deployed on prod (until 2023)
- 200+ Versions
- Zero outages
- 20+ satisfied end user
- App follows the process not vice versa



Q&A



GRACIAS

Timo Pertilä

Redes sociales y contacto



<https://bizzsummit.es>



<https://twitter.com/bizzsummities>



<http://youtube.com/bizzsummities>



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